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Microwave-assisted extraction of lignin from coconut coir using deep eutectic solvents and its valorization to aromatics

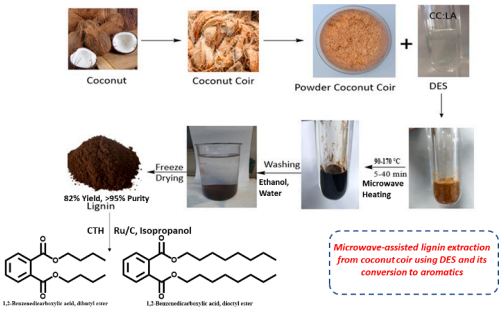
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HIGHLIGHTS

- Effective delignification of coconut coir using deep eutectic solvent was achieved.
- ChCl: LA (1:4) gave the highest lignin yield of 82% with >95% purity.
- Extraction ability of DES was correlated with Kamlet Taft parameters and pKa values.
- H factor for process was 55.5 due to efficient microwave heating at 150 °C, 20 min.
- CTH of extracted lignin gave bio-oil yield of 38.8% using Ru/C and isopropanol.

GRAPHICAL ABSTRACT



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ABSTRACT

Lignin is a rich renewable source of aromatics present in lignocellulosic biomass (LCB). The extraction of lignin from the intricate LCB network is a challenging task for successful commercialization of sustainable biorefineries. In the present study, a series of choline chloride (ChCl)-carboxylic acid based deep eutectic solvents (DESs) were used for the extraction of lignin from coconut coir under microwave irradiation. Among the synthesized DESs, ChCl: lactic acid (LA) (1:4) gave the highest lignin yield of 82% with >95% purity. Interestingly, the severity factor (H factor) for the pretreatment process was found to be a significantly lower (55.5) as compared to reported studies due to efficient microwave heating. Moreover, the DES showed good recyclability for four recycle runs thus making it a promising candidate for the delignification of LCB. Finally, the extracted lignin was converted to aromatics *via* catalytic transfer hydrogenation (CTH) using Ru/C and isopropanol as *in-situ* hydrogen donor.

1. Introduction

The last century has been evident of the over-reliance of human population over the conventional fossil fuels and environmental pollution owing to their excessive use (Mankar et al., 2021b; Modak et al., 2021b). This monotony along with the advent of growing environmental concerns and rapidly dwindling fossil reserves has led to a quest among researchers worldwide to look for sustainable sources of clean energy (Mankar et al., 2021b). Lignocellulosic biomass (LCB) is an ample and alternative renewable energy resource that can be effectively converted to obtain biofuels, biochemicals, and other value-added materials (Modak et al., 2021a). LCB, present primarily in the plant cell walls, can

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also be used as a potential substitute to the conventional fossil reserves and hence play a critical role in meeting future energy needs. LCB is a highly complex architecture wherein the cellulosic and hemi-cellulosic fractions are sandwiched between the lignin layers, overall resulting into the formation of covalently bonded strong lignin-carbohydrate complex (Mankar et al., 2021a). To achieve the maximum output of the LCB, it is essential to separate, extract, and transform chemically these biopolymers i.e., cellulose, hemicellulose, and lignin into valuable products.

Compared to other constituents in biomass, the water-insoluble aromatic polymer – lignin, is an attractive feedstock for obtaining value-added products owing to its high carbon, low oxygen content, biodegradability, and high thermal stability, among many others. However, structural complexity inherent to lignin and variability in its concentration in different LCB are major hindrance in lignin depolymerization in high quantity and high purity. Further, chemical modifications in lignin structure during pretreatment also affects its effective utilization in analysing and obtaining valuable lignols. While the current pretreatment methods have focussed largely on the extraction of cellulose and hemicellulose, separation of lignin from LCB still remains a big challenge and is therefore the key to LCB valorisation. Conventional pretreatment approaches such as organosolv, alkaline, acid, and enzymatic methods show either lower efficiency in overcoming the strong interactive energies of lignin-carbohydrate complex or lignin contamination with solvents, ultimately exhibiting lower lignin extraction potential. Ionic liquids (ILs), in this regard, have attracted significant attention due to their ability to partially or completely dissolve lignin with high extraction efficacy. However, serious concerns related to the greenness and toxicity of ILs in addition to its high viscosity and high cost render the use of ILs for LCB pretreatment (Pandey et al., 2021). Moreover, long-term recyclability and extensive postprocessing methods during ILs pretreatment also inhibit their practical applications in industrial scale-up (Mankar et al., 2021b).

Recently, the development of deep eutectic solvents (DESs) has opened up new avenues in the area of lignin extraction and its subsequent valorisation (Huang et al., 2021; Liang et al., 2021; Mankar et al., 2021c). DESs are low melting solutions obtained as a result of mixing hydrogen bond donors (HBDs) with hydrogen bond acceptors (HBAs). These neoteric eutectic mixtures resemble ILs in possessing similar physicochemical properties and are therefore also called as analogues of ILs (Chourasia et al., 2021; Pandey et al., 2021). Moreover, DESs offer significant advantages over ILs in easy of synthesis with high atom efficiency, no purification steps, insensitivity to water, non-toxicity, environmentally benign nature, and technoeconomic feasibility in their use for extraction, analytics, electrochemistry, biotechnology, catalysis, and synthesis amongst various other industrial applications. Owing to the multitude of advantages DESs exhibit over ILs, suitability of these green solvents as alternatives to ILs in pretreatment of LCB have been extensively studied. Promising results in use of DESs for lignin dissolution and extraction have been reported in earlier studies (Alvarez-Vasco et al., 2016; Liang et al., 2021; Ong et al., 2021; Tan et al., 2019; Wang et al., 2021; Xu et al., 2016).

In a recent study by Huang et al. (Huang et al., 2021), ChCl/guaiacol-based DES with $AlCl_3$ was employed for the pretreatment of wheat straw at 130 °C for 1 h giving a lignin removal of 43.3%. Similarly, Wang et al. (Wang et al., 2020) reported 69.9% of delignification of poplar at 160 °C for 3 h using lignin-based phenolic compound *p*-hydroxybenzoic acid as HBD and ChCl as HBA. Ong et al. (Ong et al., 2021) investigated the pretreatment (120 °C, 4 h) of oil palm fronds (OPF) with NaOH-DES (ChCl/urea) combined with ultrasound (60% amplitude, 30 min duration). The results showed a significantly higher lignin removal of 47% for combined pretreatment compared to 30.9% for pretreatment without ultrasound, thus indicating a synergistic effect to be responsible for improving the pretreatment efficiency. In another study, Wang et al. (Wang et al., 2021) reported the hydrothermal-DES (ChCl/lactic acid) pretreatment of moso bamboo. Under optimum conditions (140 °C, 6 h), a delignification of 80.1 wt% was achieved. Thus, the severe conditions are required for the pretreatment of biomass with longer durations (1 to 6 h) to get effective delignification. Therefore, microwave heating has proved to be an efficient pretreatment method for biomass delignification due to its fast and uniform heating at molecular level, which helps in significantly reducing the reaction time from hours to minutes (Mankar et al., 2021c). A study by Liu et al. (Liu et al., 2017) compared the conventional and microwave heating for the pretreatment of poplar using ChCl: oxalic acid DES. The results showed that it took only 3 min to get a lignin removal of 80% under microwave heating compared to 9 h required to get a similar yield under conventional heating. The use of microwave heating in combination with DES appeals to be a promising technology for the delignification of biomass from different sources which is still in its infancy. Therefore, in-depth and rigorous research is further required to explore their applicability in treating biomass to isolate lignin.

In this regard, the current study aims to use ChCl-carboxylic acid based eutectic mixtures in the treatment of coconut coir, a fibrous external portion of the fruit of *Cocus nucifera* (coconut). In India, coconut is celebrated as a symbol of prosperity, a blessing of nature, a propitious object of ceremony, a useful item of handicraft, and an essential cooking ingredient. However, coconut coir having high lignin content (~30–40 %), largely remains unused in most of these cases and hence is a source of large amounts of waste. In this study, a systematic investigation of the microwave-mediated use of different DESs in the extraction of lignin from this waste – coconut coir, in high yield and purity was done to improve the overall energy efficiency of the process and efficiency. Further studies were performed to understand the effect of pretreatment time, temperature, and HBA: HBD molar ratio on lignin extractability. DES extracted lignin was further characterized by attenuated total reflectance fourier transform infrared spectroscopy (ATR-FTIR), thermogravimetric analysis (TGA), derivative of weight loss (DTG) as a function of temperature, nuclear magnetic resonance (NMR), gel permeation chromatography (GPC) techniques. Further, the valorization of the extracted lignin to valuable aromatics was done *via* catalytic transfer hydrogenation (CTH) and the process parameters like temperature, time, and catalyst dosage were optimized to maximize the bio-oil yield. Overall, the objective of the present study is to develop a green, cost-effective, mild, and efficient process for the extraction of lignin in high purity from coconut coir and its further valorization into valuable chemicals.

2. Materials and methods

2.1. Materials

Coconut (*Cocos nucifera*) coir was procured from local vendor and air dried at 100 °C in oven for 24 h to remove the moisture. The dried coconut coir was then grinded to fine powder. The composition of cellulose, hemicellulose, and lignin present in the powdered coconut coir were calculated using standard NREL protocols as reported in our previous study (Chourasia et al., 2021). It was found that the coconut coir consisted of 28% cellulose, 15% hemicellulose, 32.8 % lignin, 2.2 % ash, and the remaining extractives. Choline chloride (ChCl; >99%), lactic acid (LA; >99%), oxalic acid dihydrate (OA; >99%), acetic acid (AA; >99%), adipic acid (AdA; >99%) butyric acid (BA; >99%), maleic acid (MA; >99%), and urea (U; >99%) were purchased from Sigma-Aldrich. 2,6-Dichloro-4-(2,4,6-triphenyl-N-pyridino)- phenolate (betaine dye 33), N, N-diethyl-4- nitroaniline (DENA), and 4-nitroaniline (NA) were purchased from Sigma-Aldrich for the calculations of Kamlet Taft solvatochromic parameters. Ru/C (5 wt% Ru loading) catalyst and isopropyl alcohol (>99%) were procured from Sigma-Aldrich. All the chemicals were used as received without any further purification.

2.2. Methods

2.2.1. Preparation of DES

A series of ChCl-based DESs were synthesized by physically mixing ChCl with different HBDs as reported in literature elsewhere (Mankar et al., 2021c, 2021b; Pandey et al., 2014). ChCl was dried overnight at 100 °C in a vacuum oven and stored in a desiccator before synthesizing the DESs owing to its hygroscopic nature. In a typical procedure, dry ChCl was weighed in a closed glass container and HBD was added to it in a particular molar ratio (Table 1) (Mankar et al., 2021c, Mankar et al., 2021d). The mixture was then heated on a magnetic plate at 100 °C with continuous stirring until a transparent liquid with no solid residue was formed. Further, the liquid was cooled down to room temperature and stored in a desiccator to keep it moisture free. The DESs selected for the extraction of lignin from coconut coir are presented in Table 1. Among all the synthesized ChCl: carboxylic acid based DESs, except maleic acid and adipic acid, all the other DESs formed a transparent eutectic mixture at room temperature (Table 1). When maleic acid and adipic acid were mixed with ChCl at 100 °C, a highly viscous white liquid with particles in suspension was formed. Moreover, when the DES was cooled at room temperature, the mixture formed white solid. Since maleic and adipic acid did not form stable DES at room temperature, these DESs were not used for further experiments.

2.2.2. Extraction of lignin from coconut coir

All the reactions for microwave-mediated extraction of lignin from coconut coir were performed in an Anton Paar microwave reactor (Model: Multiwave PRO). In a typical procedure, 0.5 g of coconut coir was mixed with 15 g DES in a quartz vessel and the reactions were performed in a temperature range of 90 to 170 °C for a reaction time of 5 to 40 min. After the reaction, the solid residue and liquid fraction were separated by vacuum filtration and the solid residue was washed with ethanol followed by water. Distilled water was then added to the liquid fraction and left overnight to precipitate lignin. The precipitated lignin was further washed several times with distilled water to ensure complete removal of DES and then freeze dried to get lignin. Further, the purity of extracted lignin was improved by washing it with ethanol: water (1:9) mixture thrice. To compare the microwave heating with conventional heating, the experiment was performed in an oil-bath at 150 °C for 8 h using 0.5 g coconut coir and 15 g DES. The % yield of the lignin extracted from coconut coir was calculated using the following equation:

$$\%\ Yield\ of\ lignin\ extracted = \frac{weight\ of\ lignin\ extracted\ from\ coconut\ coir}{weight\ of\ lignin\ in\ the\ raw\ coconut\ coir} \times 100 \quad (1)$$

For recycling of DES, all the liquid fractions were collected and further heated in a vacuum oven at 80 °C for 24 to 36 h to remove the water/ethanol thus recovering the DES. The recovered DES was used in the recycle runs (150 °C, 20 min) without any further purification. Similar recycling procedure was used for the consecutive four recycle runs.

Table 1
DESs with molar ratios of ChCl (HBA) and HBDs.

Hydrogen bond acceptor (HBA)	Hydrogen bond donor (HBD)	Molar Ratio (HBA: HBD)	Appearance (at room temperature)
ChCl	LA	1:2	Transparent liquid
ChCl	AA	1:2	Transparent liquid
ChCl	BA	1:2	Transparent liquid
ChCl	U	1:2	Transparent liquid
ChCl	OA	1:1	Transparent liquid
ChCl	MA	1:1	White solid
ChCl	AdA	1:1	White solid

2.2.3. Depolymerization of extracted lignin by catalytic transfer hydrogenation

The experiments for the catalytic transfer hydrogenation (CTH) of extracted lignin were carried out in an Anton Paar microwave reactor (Model: Monowave 300). In a typical procedure, the borosilicate glass vial of the reactor was charged with 30 mg extracted lignin, Ru/C catalyst (1.5 to 9 mg), 5 mL isopropyl alcohol, and a magnetic bead was placed in the vial. After that, the vial was placed in the microwave reactor and the reactions were performed in a temperature range of 80 to 180 °C for a reaction time of 10 to 60 min. When the reaction was completed, the mixture was cooled to 50 °C using the cooling system provided with the reactor. Then, the resulting product mixture was centrifuged at 12000 rpm for 15 min to separate the solid residue from the liquid fraction. Further, the solid residue, which consisted of unreacted lignin and catalyst, was washed with isopropyl alcohol thrice to ensure complete separation of heavy bio-oil from the solid residue. Then, the solid residue was dried in vacuum oven overnight at 100 °C and weighed. The liquid fraction obtained after centrifugation and washing of solid residue with isopropyl alcohol were collected together and dried in a vacuum oven overnight at 75 °C to remove isopropyl alcohol. The resulting liquid fraction was weighed and denoted as bio-oil. The % yield of solid residue and bio-oil was calculated using the following equations:

$$\%\ Yield\ of\ bio-oil = \frac{weight\ of\ bio-oil}{initial\ weight\ of\ lignin\ charged} \times 100 \quad (2)$$

$$\%\ Yield\ of\ solid\ residue = \frac{weight\ of\ solid\ residue}{initial\ weight\ of\ lignin\ charged} \times 100 \quad (3)$$

2.2.4. Characterization of extracted lignin, coconut coir, and bio-oil

The lignin extracted from coconut coir using ChCl:LA DES, Kraft lignin, untreated raw coconut coir, and the recovered coconut coir after the pre-treatment were characterized using ATR-FTIR spectroscopy technique. Thermo Scientific Nicolet iS50 FTIR spectrometer was used to record the ATR-FTIR spectra in a range of 1000 to 4000 cm^{-1} with 64 scans. The samples were dried in vacuum oven at 100 °C for 24 h before the analysis.

Thermal stability of lignin extracted from pretreatment of coconut coir using ChCl: LA DES was determined using TGA (SDT Q600) and compared with the Kraft lignin. The TGA analysis was done in the presence of nitrogen flow (100 mL/min) and the sample was heated from room temperature up to 800 °C at a ramp rate of 10 °C/min. Further, the DTG as a function of temperature (%weight loss/°C) of the lignin samples was also determined.

The proton (^{1}H) and carbon (^{13}C) NMR spectra were recorded for the extracted lignin and compared with the Kraft lignin. NMR was recorded on Bruker Biospin ADVANCE (500 MHz) and the samples were dissolved in DMSO-d_6 solvent for the NMR acquisition.

The molecular weight distribution of the lignin extracted from coconut coir after DES pretreatment was measured using GPC technique. The lignin sample was dissolved in THF solvent at room temperature and then injected in the GPC instrument (Perkin-Elmer Series-200) equipped with RI detector.

The effect of ChCl: LA DES pretreatment on the morphology of coconut coir was studied by field emission scanning electron microscope (FESEM). JEOL JMS-7800F prime was used to take the FESEM images at different magnifications (1500x, 2000x, 5000x). Prior to FESEM analysis, the samples were dried overnight at 100 °C in a vacuum oven. A conductive carbon tape was used to mount the samples on an aluminium stub and coated with platinum under vacuum.

Stock solutions of the Kamlet-Taft dyes namely betaine dye 33 (BD33), 4-nitroaniline (NA), and N, N-diethyl-4- nitroaniline (DENA) were prepared by dissolving the dyes in ethanol and the prepared solutions were stored in amber vials at 4 °C. Then, the stock solution was carefully transferred in 1 cm^2 quartz cuvette for recording the UV-Vis spectra. Further, the ethanol was slowly evaporated using a stream of pure nitrogen gas. The DES was then directly added to the cuvette and the solution was sonicated to ensure proper uniform mixing. Finally, the absorption spectra were recorded using Agilent Cary Series UV-Vis spectrophotometer. All the measurements were done in triplicates to ensure reproducibility. The Kamlet Taft solvatochromic parameters (α, β, and π*) for the synthesized DESs are given in Table 2.

The following equations were used for the calculation of α, β, π* of the synthesized DESs:

$$ET\ (33) = \frac{28591}{\lambda maxBD33} \tag{4}$$

$$ET\ (30) = 0.9953\ \times ET\ (33) - 8.1132 \tag{5}$$

$$\beta = \frac{\left(1.035\ \times \frac{10^4}{\lambda maxDENA}\right) + 2640 - \left(10^4/\lambda maxNA\right)}{2800} \tag{6}$$

$$\pi^*\ = 0.314\ \times\ \left(27.52 - 10^4/\lambda maxDENA\right) \tag{7}$$

$$\alpha = 0.00649\ \times\ ET\ (30) - 2.03 - 0.72\ \times\ \pi^* \tag{8}$$

The identification of the products present in the bio-oil was done using Shimadzu GCMS-QP2010 Plus equipped with Rtx-5 MS fused silica capillary column (30 m ×0.25 mm ×0.25 μm) with helium as a carrier gas. The program started at 50 °C and then increased with a ramp rate of 10 °C/min to 120 °C with a hold time of 12 min. Further, the temperature was increased to 280 °C with a ramp rate of 10 °C/min and held for 8 min. Finally, the temperature was raised to 300 °C with a ramp rate of 10 °C/min and then held for 2 min.

3. Results and discussion

3.1. Pretreatment of coconut coir with DESs for lignin extraction

3.1.1. Screening of DESs for lignin extraction

A series of ChCl-based DESs (Table 1) were used for the extraction of lignin from coconut coir using microwave reactor. All the reactions were performed in the presence of coconut coir and DES in a mass ratio of 1:30 at a temperature of 125 °C for 20 min. The effect of different HBDs in the ChCl-based DESs, on the % yield of lignin extracted was studied, and the results are shown in Fig. 1a. Kamlet-Taft solvatochromic parameters α, β, and π* which represents the hydrogen-bond acidity, hydrogen-bond basicity, and polarizability/dipolarity of the DESs, respectively, were calculated based on our previous studies (Pandey et al., 2014; Pandey and Pandey, 2014). These parameters have been previously reported to interpret the relation between the DESs properties and their ability to extract lignin from biomass (Liang et al., 2021; Liu et al., 2017). When the lignin extraction was done using ChCl: urea, negligible lignin extraction (<1%) was achieved, indicating that the absence of acidity in urea might be responsible for the low lignin yield. However, a significant extraction of lignin from the coconut coir was observed when the pretreatment were performed using ChCl: carboxylic acid based DESs. The hydrogen-bond acidity (α) of the DES is previously reported to govern the cleavage of lignin-carbohydrate-complex (LCC) in the biomass (Liang et al., 2021).

Table 2
Kamlet Taft solvatochromic parameters (α, β, and π*) for the synthesized DESs.

DES	α	β	π*
ChCl: OA	1.32	0.62	1.09
ChCl: LA	1.17	0.51	1.12
ChCl: AA	0.92	0.54	1.1
ChCl: BA	0.81	0.53	0.96

ChCl: OA DES gave the highest lignin extraction of 37.8% as it had the highest α value of 1.32. Further, when ChCl: LA DES was used for the pretreatment of coconut coir, a lignin extraction of 35.3% was observed. A slight decrease in the lignin extraction compared to ChCl: OA DES can be related to the lower α value (1.17). Similar trend of results was obtained for ChCl: AA and ChCl: BA DES based on the α values (Fig. 1a). The effect of aliphatic chain length of the HBD on the lignin extraction ability was also studied by comparing the ChCl: AA and ChCl: BA DES. When ChCl: AA DES was used for the pre-treatment of coconut coir, a lignin yield of 15.2% was achieved. However, when the ChCl: BA DES was applied for the pre-treatment, the lignin yield decreased to half (7.6%). The results indicated that increase in alkyl chain length of the HBD reduced the ability of that particular DES to extract lignin. The electron density of the oxygen in the hydroxyl group of carboxylic acid increases with the increase in chain length of the electron donating alkyl group. As a result, the strength of the hydrogen bond between the oxygen and hydrogen increases thus limiting the ability to donate proton. Therefore, the DES with shorter chain length showed better activity for lignin removal as it improved the solvent-solute interaction by easily donating the proton (Teles et al., 2017).

It is also noteworthy to mention here that the pKa value of the HBDs in the DES also plays an important role in indicating the lignin extraction ability of a particular DES. The results of lignin extraction from coconut coir are in agreement with the pKa values, e.g., ChCl: AA (pKa = 4.76) showed better lignin yield compared to ChCl: BA (pKa = 4.82). In case of ChCl: LA DES, higher lignin extraction yield of 35.3% was achieved because of the lower pKa value (3.86). However, despite slightly higher lignin yield of 37.8% in case of ChCl: OA DES owing to the higher acidity (pKa_1 = 1.46, pKa_2 = 4.4), a significant charring of the coconut coir recovered after the pre-treatment, was observed. High acidity of ChCl: OA can be attributed for the charring of the coconut coir. Moreover, release of CO_2 is also reported due to the activation of carboxyl groups in the OA. However, it is desired to recover the lignin free coconut coir after the pretreatment so that it can be further utilized to produce fermentable sugars *via* enzymatic hydrolysis which can further be converted to bioethanol. Therefore, ChCl: LA DES was used for further reactions as the solid (coconut coir) was recovered compared to the charred solid obtained in case of ChCl: OA DES.

3.1.2. Effect of reaction temperature and time on lignin extraction

The effect of reaction temperature on the extraction of lignin using ChCl: LA DES was also studied. Reactions were performed in the presence of coconut coir: DES (1:30) at 90 °C, 110 °C, 130 °C, 150 °C, 170 °C for 20 min (Fig. 1b). When the reaction was done at 90 °C, a lignin yield of 10.9% was achieved. Further increasing the temperature led to a moderate increase in the amount of lignin extracted and a yield of 28.5%, 35.1% was obtained for 110 °C, 130 °C temperatures, respectively. Interestingly, lignin yield increased significantly to 73.9% when the temperature was further increased to 150 °C. The results indicated that the higher reaction temperatures favoured the extraction of lignin. An increase in the temperature leads to the reduction in viscosity of the DESs (Mankar et al., 2021c). As a result, the mass transfer limitations are reduced resulting in improved interactions between the DES and the coconut coir. However, when the reaction temperature was further increased to 170 °C, a slight decrease in yield of lignin to 71.2% was observed. Similar results showing a drop in the lignin yield at higher

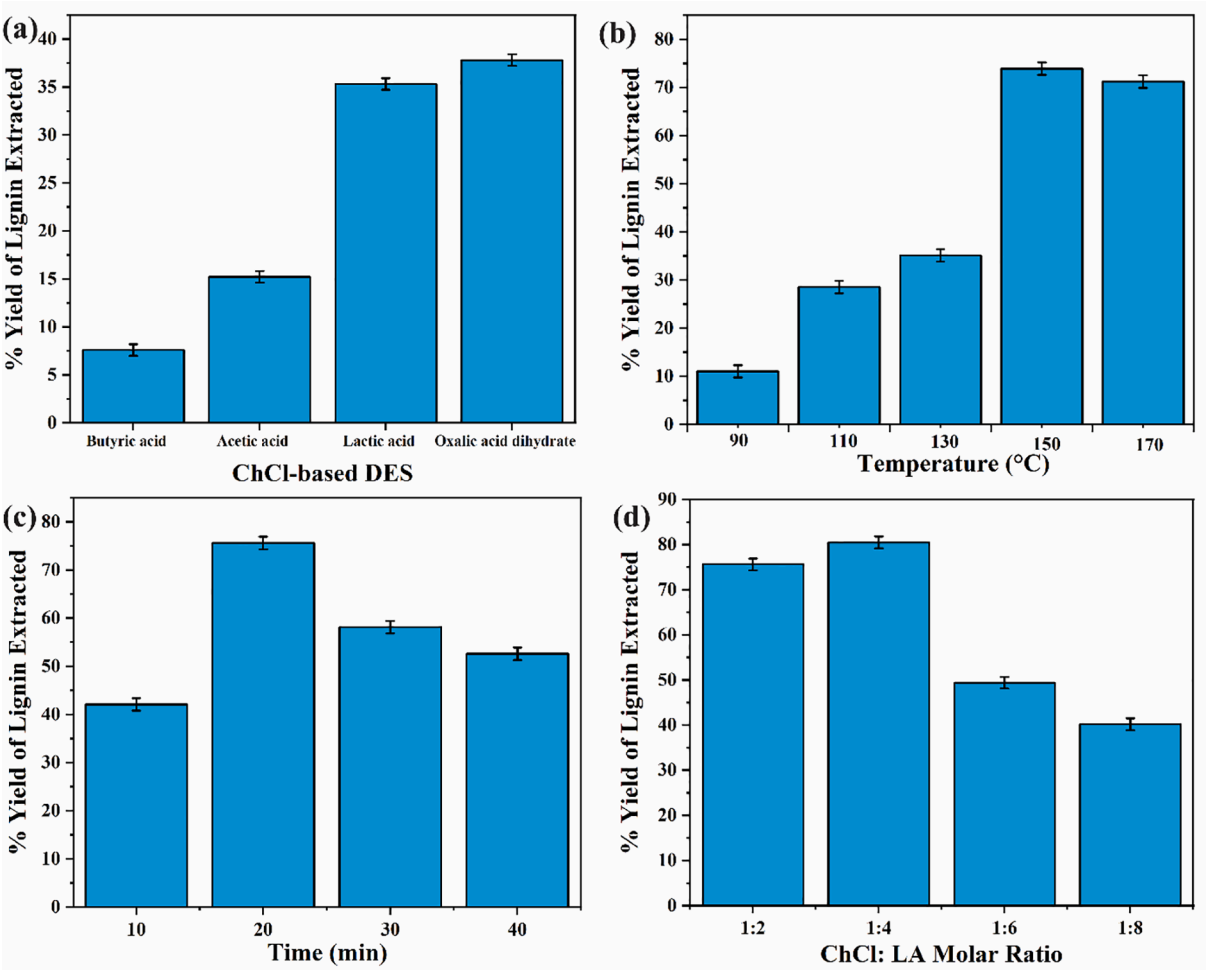


Fig.1. (a) Screening of ChCl-based DESs with different HBDs for lignin extraction, (b) Effect of temperature on the lignin extraction from coconut coir using ChCl: LA DES, (c) Effect of time on the lignin extraction from coconut coir using ChCl: LA DES, (d) Effect of ChCl: LA molar ratio on the lignin extraction from coconut coir using ChCl: LA DES.

temperatures (>145 °C) were reported in the previous studies and the reason for the reduction in extraction efficiency might be the redeposition of lignin on the residual solid after the pre-treatment process (Alvarez-Vasco et al., 2016; Lou et al., 2019). Moreover, the degradation of the carbohydrates might as well take place at elevated temperatures thereby reducing the recovery of lignin free cellulose-rich coconut coir (Alvarez-Vasco et al., 2016; Lou et al., 2019). Thus, the results suggested that an optimum reaction temperature is required to extract the lignin from coconut coir with maximum yield without compromising the recovery of cellulose-rich coconut coir. Therefore, all the further reactions were done at 150 °C as the highest amount of lignin was extracted at this temperature.

The lignin extraction reactions were performed at 150 °C between time interval of 5 to 40 min to understand the effect of time on the lignin extraction and the results are as shown in Fig. 1c. Initially when the reactions were performed for 5 and 10 min in the microwave reactor, a lignin yield of 35.1% and 42.4%, respectively were achieved. When the reaction time was further increased to 20 min, the rate of lignin extraction increased significantly giving a high lignin yield of 73.9%. However, when the experiments were performed for longer time intervals (>20 min), a decrease in lignin yield was observed. The prolonged interactions of DES with the coconut coir might cause changes in the structural properties of the lignin which results in redeposition of lignin on the surface of the residual solid (Lou et al., 2019). Therefore, a reaction time of 20 min was found to be optimum for lignin extraction with high yields.

Table 3
H factor for the pre-treatment of coconut coir using DES under different reaction conditions.

Time (min/h)	Temperature (°C/K)	H factor
20/0.33	90/363.15	0.1
20/0.33	110/383.15	1.04
20/0.33	130/403.15	8.4
20/0.33	150/423.15	55.5
20/0.33	170/443.15	310
5/0.083	150/423.15	13.8
10/0.16	150/423.15	27.7
30/0.5	150/423.15	83.3
40/0.66	150/423.15	111.1

H-factor, a parameter involving the time and temperature to determine the severity of a particular chemical pulping pretreatment process for lignin removal, was also determined in this study (Table 3). H-factor is a single variable model which is used to determine the rate of delignification with an assumption that the delignification is one single reaction (Alvarez-Vasco et al., 2016). It is calculated using the following equation:

$$H = \int_0^t exp\left(43.2 - \frac{16115}{T}\right) dt \quad (9)$$

where t is the time in hour (h) and T is the temperature in kelvin (K)

In the present study, the H factor for the different reaction conditions is in a range of 0.1 to 310. Among all the experiments, highest lignin extraction yield of 73.9% was observed at 150 °C for a pre-treatment of 20 min, which had a significantly lower H factor of 55.5. Compared to the present study, the previous reports showed a lignin extraction yield of 58.2% (D. fir wood), 78.5% (poplar wood) with a H factor of 943 (145 °C, 9 h), 629 (145 °C, 6 h), respectively (Alvarez-Vasco et al., 2016). The results clearly indicate that the pre-treatment severity in the present study is significantly lower compared to the previous studies, thus making microwave-mediated heating an interesting choice for the pretreatment of LCB.

3.1.3. Effect of molar ratio of ChCl to LA on lignin extraction

To understand the role of ChCl and LA in the extraction of lignin, experiments were performed in the presence of ChCl: LA DES with different molar ratios (1:2, 1:4, 1:6, 1:8). When the experiment was performed in the presence of pure LA at 150 °C for 20 min, a low lignin yield of 16.9% was observed. However, when the extraction run was performed in the presence of ChCl: LA (1:2), a significant increase in the lignin yield from 16.9% (in the absence of ChCl) to 73.9% was achieved. The results indicated that ChCl played a synergistic effect with LA in the extraction of lignin from the coconut coir, thus resulting in higher lignin yield. The delignification of coconut coir was initiated *via* the cleavage of LCC. This involved weakening of strong hydrogen bonding present within coconut coir due to the formation of hydrogen bonds between the hydroxyl group of lignin/carbohydrate and chloride ions of ChCl: LA DES (Liu et al., 2017).

Additionally, the role of LA in the extraction of lignin was also accessed. As such, the molar ratio of LA in ChCl: LA DES was increased from 1:2 to 1:4. A two-fold increase in LA concentration in the DES resulted in enhanced lignin yield, i.e., the lignin yield was found to be 73.9% and 82% for 1:2 and 1:4 ratio of ChCl: LA, respectively. The results indicated that the increase in LA content in DES has promoted the extraction ability of DES, thus improving the overall yield of the extracted lignin. The reason for the improved delignification efficiency was the promoted cleavage of LCC due to the increased hydrogen-bond acidity associated with the LA content (Chen et al., 2019). However, when the reactions were performed at higher ChCl: LA molar ratios (>1:4), the lignin yield was found to decrease as shown in Fig. 1d. Overall, the results indicated that both LA and ChCl plays an equally important role in the extraction of lignin and an optimum molar ratio is required to get the maximum extraction efficiency. In the present study, a ChCl: LA with a molar ratio of 1:4 was found to be the optimum DES as it gave the highest yield of extracted lignin (82%).

3.1.4. Comparison of conventional and microwave heating and DES recyclability

The effect of mode of heating on the efficiency of lignin extraction from coconut coir was also compared. Experiment was conducted in an oil-bath with 0.5 g coconut coir and 15 g ChCl: LA (1:4) DES for lignin extraction under conventional heating. When the reaction was performed at 150 °C for 20 min, only 13.5% lignin was extracted. Under similar reaction conditions, when the reaction was performed under microwave heating, the lignin extraction was 6 times higher (82%) compared to conventional heating. To further improve the lignin extraction under conventional heating, the reaction time was extended to 8 h which resulted in only 46% of lignin extraction. Besides, the H factor for the conventional heating (150 °C, 8 h) was also significantly higher (1334.1) compared to the significantly lower H factor (55.5) for microwave heating (150 °C, 20 min), while the lignin extraction was also approximately half of that for microwave heating. Lower lignin extraction ability using DES under severe reaction conditions in case of conventional heating is a result of heating *via* conduction which involves heat losses due to temperature gradients. On the contrary, an electromagnetic field is applied in case of microwave heating which facilitates vibration and rotation of the polar molecules present in LCB, hence releasing heat at faster rate (Mankar et al., 2021c). This prevents any heat losses during microwave heating and ensures uniform heating. Moreover, the microwave radiation helps in increasing the molecular polarity and also maximizes the ionic characteristics of DES, resulting in better lignin extraction at lower reaction time (Liu et al., 2017). It is reported for microwave heating that the polar groups present in the complex LCB structure can further create hot spots and led to an explosion effect within the particles. As a result, the disruption of LCB complex takes places resulting in enhanced lignin extraction. Overall, the microwave heating was highly effective in extracting lignin from coconut coir in high yields using ChCl: LA DES compared to the conventional heating under milder reaction conditions.

The recyclability of DES is an important parameter in terms of development of a sustainable future biorefinery. In this study, we have recycled the DES and reused it for the extraction of lignin from coconut coir to check its recyclability. All the recycle runs were performed under the optimized reaction conditions (150 °C, 20 min, ChCl: LA (1:4) DES). The DES showed considerable lignin extraction in the recycle runs. When the DES was recycled after the first run, the lignin yield was found to be slightly reduced from 82% (fresh run) to 78.1%. Additionally, a lignin yield of 75.5% and 71.8% was achieved for second and third recycle run, respectively. However, a further gradual decrease in the lignin yield to 68.5 % was seen after the fourth cycle. The probable reason for the reduction in the extraction ability might be the loss of some amount of DES during the workup procedures which might have reduced the overall acidity of the DES reaction system.

In earlier reported literatures, Wu et al. (Wu et al., 2014) studied the organosolv pretreatment of hybrid poplar without catalyst using 80% aqueous solution of methanol at 220 °C for 30 min and a lignin yield of 45.5% was reported. Further, to improve the lignin yield under milder reaction conditions, some studies have reported the use of acid/alkali catalysts. Zhong et al. (Zhong et al., 2021) investigated effect of alkali catalyst (NaOH) on the organosolv pretreatment (methanol/water (65/35 v/v)). A lignin yield of 64.3% was reported at 80 °C, 1 h using 2 wt.% NaOH dosage. In another study, Mondylaksita et al. (Mondylaksita et al., 2020) reported the acid-catalyzed (0.07% H_2SO_4) organosolv pretreatment (ethanol) of Oil palm empty fruit bunch. Under the pretreatment conditions of 210 °C and 1.5 h, a lignin yield of 64.9% was reported. In a similar study by Chen et al. (Chen et al., 2021), a ternary solvent system (acetone/phenoxyethanol/water) was utilized for the pretreatment of Amorpha using H_2SO_4 as catalyst. Under optimum conditions (130 °C, 70 min, 0.15 M H_2SO_4) a lignin yield of 35.6% was reported. Wei et al. (Wei et al., 2021) compared the acid and alkali catalyzed pretreatment of bagasse using organosolv approach (ethylene glycol/water (9/1 v/v)). When the pretreatment was done at 130 °C for 1 h, a lignin yield of 39.2% and 47.9% were reported for ethylene glycol/H_2O-HCl and ethylene glycol/H_2O-NaOH, respectively. The major drawback of organosolv process is volatility, flammability, high cost, and solvent recovery. Moreover, recyclability of the homogenous acid and alkali catalyst is also a major challenge. In addition, the acid pretreatment involves the problems like corrosion and formation of fermentation inhibitors while the alkali pretreatment is mainly associated with high costs for neutralization of slurry (Mankar et al., 2021c). Thus, our study is advantageous over these reported established methods in terms on cost-effectiveness, biodegradability, ease of synthesis, and recyclability of the DESs.

3.2. Characterization of extracted lignin and raw coconut coir

3.2.1. ATR-FTIR

ATR-FTIR spectra of the lignin extracted from coconut coir using ChCl: LA DES was recorded and compared with the spectra for Kraft lignin (see e-supplementary material). All the spectra showed a broad absorption band at 3415 cm^{-1} corresponding to the O—H stretching vibration of aliphatic and phenolic OH groups present in lignin (Yan et al., 2015; Zhao et al., 2014). Moreover, the absorption bands at

wavelength 2925 and 2860 cm^{-1} corresponds to the alkane C—H stretching vibrations in methyl, methylene groups (Yan et al., 2015; Zhao et al., 2014). Absorption bands at 1606, 1507, 1455 cm^{-1} corresponds to the C═O stretching conjugated to aromatic ring, aromatic skeletal vibrations, C—H bending in the methyl group, respectively (Yan et al., 2015; Zhao et al., 2014). The peak at wavelength 1264 cm^{-1} was assigned to the C-O stretch in the guaiacol rings while the peak at 1220 cm^{-1} corresponded to the C-O stretch in the syringyl rings (Zhao et al., 2014). Absorption peak at 1125, 1030 cm^{-1} corresponded to the C-O deformation in the primary and secondary alcohols or aliphatic esters (Zhao et al., 2014). Thus, it was confirmed that the extracted lignin had all the functional groups corresponding to the native lignin structure.

To further confirm the removal of lignin from the coconut coir, we also did the ATR-FTIR analysis of coconut coir recovered after the pretreatment with ChCl: LA DES and compared it with the untreated raw coconut coir (see e-supplementary material). As expected, the reduction in peak intensities corresponding to lignin were observed in case of pretreated coconut coir compared to the untreated raw coconut coir. The absorption bands at 1610, 1510, 1440 cm^{-1} corresponding to the aromatic skeletal vibration in lignin showed reduction in peak intensity (Yan et al., 2015; Zhao et al., 2014). Moreover, the absorption bands at 1264, 1220 cm^{-1} showed reduced peak intensity for the pretreated coconut coir and the peaks corresponded to C-O stretch in guaiacol, syringyl rings, respectively (Zhao et al., 2014). Similar reduction in the absorption peak intensity was also found for peaks corresponding to 1125, 1030 cm^{-1} representing the C-O deformation in the primary and secondary alcohols or aliphatic esters (Zhao et al., 2014). The absorption band at 820 cm^{-1} corresponding to C—H bending showed a reduction in peak intensity. Thus, the results confirmed the removal of lignin from coconut coir after the pretreatment with ChCl: LA DES.

3.2.2. TGA

Thermal stability of the lignin was studied using TGA analysis (see e-supplementary material). The initial weight loss between 25 and 155 °C corresponded to the removal of moisture and volatiles with low molecular weight (Rashid et al., 2016). The moisture content of Kraft lignin was significantly higher (7.9 wt.%) compared to the extracted lignin (4.1 wt.%) as can be seen from the TGA curves. Second major weight loss occurred between 156 and 550 °C which is associated with the fragmentation of several inter-unit linkages (C—O—C, C—C) giving products like phenolics, gases, etc (Rashid et al., 2016). Moreover, it is reported that the cracking of methoxy groups in the lignin leads to the release of methane gas while CO gas is released due to C═O, C—O—C cleavage in a temperature range of 400 to 700 °C. Further, at higher temperatures (551 to 800 °C), aromatic rings undergo thermal decomposition producing char or coke (Rashid et al., 2016). From the TGA plots, it can be clearly seen that the lignin is not completely degraded even at 800 °C due to the presence of high-condensed residual char (Schorr et al., 2014). The Kraft lignin showed a residual char of 47.8% while the DES extracted lignin had a residual charge of 33.4%, indicating the presence of more condensed C—C linkages in case of Kraft lignin compared to the extracted lignin (Schorr et al., 2014). The lower number of C—C linkages in the extracted lignin might help in its easier valorization to aromatics.

3.2.3. DTG

The DTG analysis was also performed and the DTG curves (see e-supplementary material) showed a peak at low temperature (50 to 150 °C) related to the removal of water (Moustaqim et al., 2018). The maximum weight loss rate (wt.%/°C) of 0.08 was seen for Kraft lignin which was four times compared to the extracted lignin (0.02), thus indicating more moisture content in the Kraft lignin. Further, a sharp maxima was recorded at 315 °C which showed a maximum weight loss of 0.2 for Kraft lignin and 0.14 for extracted lignin due to the cleavage of β-O-4 linkage (Kim et al., 2013). The shoulder peak at higher temperatures (450 to 550 °C) corresponded to the cleavage of C—C bond in the lignin (Kim et al., 2013). The peak at around 700 °C corresponded to the formation of condensed lignin structure (char) *via* condensation/polymerization reactions occurring at higher temperatures (Moustaqim et al., 2018; Schorr et al., 2014). The DTG curves shows that the Kraft lignin had a weight loss rate (0.04) which was twice compared to the extracted lignin (0.02), thus indicating the presence of more C—C condensed structural bonds in case of Kraft lignin (Moustaqim et al., 2018; Schorr et al., 2014).

3.2.4. ^{1}H NMR

The chemical structure of the extracted lignin and the associated functional groups were further characterized using ^{1}H NMR spectroscopy (see e-supplementary material). The peaks between 0.8 and 2 ppm corresponded to the aliphatic hydrocarbon region mainly consisting of primary, secondary, and tertiary alkyl groups (Deepa and Dhepe, 2015; Li et al., 2020; Rashid et al., 2016). The peaks between 1.4 and 2.2 ppm corresponds to the proton of aliphatic acetates while the signals at 2.2 to 2.5 ppm represents the proton of aromatic acetates (Rashid et al., 2016). A sharp peak at 2.5 to 2.7 ppm corresponds to the DMSO-d_6 solvent while peak between 3.5 and 3.7 ppm represents protons in water in DMSO-d_6 (Rashid et al., 2016; Zhang et al., 2010). Methoxy protons in the lignin are represented by the peaks between 3.5 and 4.1 ppm (Li et al., 2020; Rashid et al., 2016). The peaks corresponding to the presence of aromatic protons in lignin appear between 6.5 and 8.5 ppm (Deepa and Dhepe, 2015). The signals corresponding to the aromatic protons in the syringyl (S) propane showed peaks between 6.1 and 6.7 ppm. Further, the aromatic protons in the guaiacyl (G) propane and *p*-hydroxyphenyl propane units showed peaks between 6.7 and 7.5 ppm and 7.4 to 8 ppm, respectively (Rashid et al., 2016; Zhang et al., 2010). Overall, most of the functional groups corresponding to the lignin structure were preserved and present in the extracted lignin indicating the DES extraction process has not much altered the native structure of the lignin preserving the majority of functional groups.

3.2.5. ^{13}C NMR

The extracted lignin was further analysed using ^{13}C NMR technique (see e-supplementary material). The peaks between 20 and 40 ppm are assigned to aliphatic side chain region of the lignin (Deepa and Dhepe, 2015). The peaks corresponding to the CH_3O species are present in a range of 54 to 57 ppm (Yue et al., 2020). Moreover, the peaks between 55 and 90 ppm corresponded to the sp^3 carbon next to oxygen and the peaks between 110 and 150 ppm represents the sp^2 carbons (Deepa and Dhepe, 2015). The carbon in the ester group of the lignin structure are assigned to the peaks between 175 and 180 ppm (Deepa and Dhepe, 2015). Furthermore, no signals were observed between 90 and 102 ppm which confirmed the absence of carbohydrates in the extracted lignin fraction. Interestingly, lack of signals between 72 and 90 ppm confirmed the absence of ether linkages in the lignin structure. Thus, the results suggested that the acidic DES extracted the lignin from the coconut coir *via* cleavage of the β-O-4 ether linkages (Alvarez-Vasco et al., 2016).

3.2.6. GPC

The weight average molecular weight (M_w) and number average molecular weight (M_n) of the extracted lignin were determined using GPC analysis. The values of M_w and M_n were 1260 g/mol and 407 g/mol, respectively, for the lignin extracted using ChCl:LA DES. Based on a previous study, the lignin extracted from coconut coir using milling technique i.e., milled-wood lignin (MWL) had a M_w and M_n values of 7900 g/mol and 2900 g/mol, respectively (Rencoret et al., 2013). Notably, these values are significantly higher compared to the lignin extracted in the present study. The probable reason for the decrease in M_w, M_n might be the depolymerization of lignin to form low-molecular weight lignin fragments due to the acidic DES pretreatment, which cleaved the intermolecular β-O-4 linkages (confirmed by TGA/DTG, NMR) without causing the condensation reactions (Alvarez-Vasco et al., 2016; Kim et al., 2013; Xie et al., 2021). Overall, the extracted lignin will

have better polymer stability due to the high probability of disruption of easily cleavable β-O-4 ether bonds compared to the stronger C—C bonds (Alvarez-Vasco et al., 2016).

3.2.7. FESEM

The morphological changes in the structure of coconut coir after the pre-treatment with ChCl: LA DES was studied using FESEM technique. The FESEM images were taken for the untreated raw coconut coir at different magnifications and compared with the coconut coir recovered after the pretreatment with ChCl: LA DES to confirm the removal of lignin (see e-supplementary material). The untreated coconut coir showed smooth surface which indicated that the fibres were tightly packed and had an ordered structure. However, when the coconut coir was pretreated with ChCl: LA DES, the structure of the coconut coir collapsed and the surface showed cracks thus loosening the fibrous structure (Du et al., 2016). The disruption of the structure after the pre-treatment indicated the removal of lignin due to the cleavage of the LCC network (Du et al., 2016). As a result, the internal fibrous structure was exposed after the pre-treatment. Thus, the lignin free coconut coir obtained after the pretreatment with high surface area can further be utilized for enzymatic hydrolysis to produce fermentable sugars, which can be further converted to bio-ethanol in high yields.

3.3. Optimization of reaction parameters for catalytic transfer hydrogenation (CTH)

The lignin extracted from the coconut coir using ChCl: LA DES was further valorised by employing Ru/C catalyst for CTH in the presence of isopropyl alcohol. Effect of different reaction parameters on the lignin derived bio-oil yield was studied and the results are presented in Table 4. Temperature plays a critical role in the CTH reactions. Therefore, reactions were performed in a temperature range of 80 to 180 °C for 20 min in a microwave reactor to study the effect of temperature on the bio-oil yield. When the reaction was done at 80 °C, a bio-oil yield of 28.5% was achieved. Further, when the reactions were carried out at elevated temperatures of 100 °C and 120 °C, the yield of bio-oil increased to 32.7% and 38.8%, respectively. The results suggested that the depolymerization of lignin is temperature dependent, i.e., increases with increase in temperature and vice-versa. However, when the reaction temperature was further increased (>120 °C), a reduction in the bio-oil yield was observed (Table 4), with a lowest bio-oil yield of 25.8% at 180 °C. The reason for the decrease in the bio-oil yield is the decomposition or condensation of the lignin derived products, which might lead to char formation at higher temperatures (Hita et al., 2018; Long et al., 2015). Therefore, 120 °C was considered as the optimum reaction temperature as highest bio-oil yield was achieved at this temperature.

The reaction time is another crucial parameter which governs the bio-oil yield in the CTH reactions. Therefore, a series of reactions were performed at 120 °C and the reaction time was varied between 10 and 60 min. When the reaction was conducted for 10 min, a bio-oil yield of 29.9% was achieved. Moreover, further increasing the reaction time to 20 min led to improved bio-oil yield of 38.8%. But, when the reaction time was >20 min, a decline in the bio-oil yield was observed (Table 4) and only 26.1% yield was achieved at 60 min. The probable reason for the decreasing trend in the bio-oil yield at longer reaction time might be the re-polymerization/re-condensation of the reactive intermediates which may lead to humin/char formation as confirmed from the increased solid residue (Long et al., 2015). Therefore, further reactions were performed for a reaction time of 20 min as maximum bio-oil yield was achieved at this time.

Another important parameter which affects the yield of bio-oil in the CTH of lignin is the catalyst dosage. The catalyst plays an important role in cleavage of the linkages in the lignin to produce desired valuable aromatic compounds. When the reactions were performed at 120 °C for 20 min at 1.5 mg Ru/C catalyst, a bio-oil yield of 25.3% was observed. Further, the bio-oil yield increased to 27.2% and 32.1% when the catalyst dosage was increased to 3 mg and 4.5 mg, respectively. The results indicated that higher catalyst dosage (up to 4.5 mg) provided more active sites for the CTH reaction leading to more cleavage of linkages in the lignin structure, thus improving the bio-oil yield. However, when the catalyst dosage was further increased (>4.5 mg), a decline in the bio-oil yield was observed (Table 4). The reason for the reduced yield is the repolymerization of the reactive intermediates to form char at higher catalyst dosage (Hita et al., 2018). Under optimum conditions, the GC–MS results (see e-supplementary material) showed a highly selective formation of two major products namely, 1,2-Benzenedicarboxylic acid, dibutyl ester (48%) and 1,2-Benzenedicarboxylic acid, dioctyl ester (47.5%), which are commonly used as plasticizers. These compounds are formed *via* cleavage of the linkages present in the DES extracted lignin *via* hydrogenation/hydrogenolysis reactions. In addition, minor products like 1,2-Benzenedicarboxylic acid, butyl diethylhexyl ester (1.7%) and hexanedioic acid, bis (2-ethylhexyl) ester (2.8%) were also detected. The formation of aliphatic long chain fatty acid ester is reported earlier, which are formed because the reactive sites (C═O or C—O bonds) within the low molecular weight compounds undergoes re-condensation reaction (Kim et al., 2015).

Table 4
Effect of reaction parameters on CTH of lignin extracted using ChCl: LA DES from coconut coir.

Extracted Lignin (mg)	Ru/C (mg)	Isopropyl alcohol (mL)	Temperature (°C)	Time (min)	% Yield of bio-oil	% Yield of solid residue
30	4.5	5	80	20	28.5	71.5
30	4.5	5	100	20	32.7	67.3
30	4.5	5	120	20	38.8	61.2
30	4.5	5	140	20	32.3	67.7
30	4.5	5	160	20	30.1	69.9
30	4.5	5	180	20	25.8	74.2
30	4.5	5	120	10	29.9	70.1
30	4.5	5	120	30	31.3	68.7
30	4.5	5	120	40	29.7	70.3
30	4.5	5	120	50	28.6	71.4
30	4.5	5	120	60	26.1	73.9
30	1.5	5	120	20	25.3	74.7
30	3	5	120	20	27.2	72.8
30	4.5	5	120	20	32.1	67.9
30	6	5	120	20	29.5	70.5
30	7.5	5	120	20	28	72

4. Conclusions

A practical approach towards effective pretreatment of coconut coir using DESs to extract high purity lignin is demonstrated in this study. Under optimum conditions (150 °C, 20 min, ChCl: LA DES), a lignin yield of 82% was achieved using microwave heating. The ability of DES to extract lignin was directly correlated with their Kamlet Taft parameters and pKa values. The extracted lignin had all the functional groups corresponding to the native lignin and a purity of >95%. Further, the CTH of extracted lignin gave a bio-oil yield of 38.8% under mild reaction conditions (120 °C, 20 min) using Ru/C catalyst.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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